

This session we're going to cover:

- ascii/char
- number guesser mini project

ASCII (استاندارد آمریکایی برای تبادل اطلاعات)

یک استاندارد کدگذاری نویسه ۷-بیتی است که ۱۲۸ نویسه را نمایش می‌دهد
شامل حروف انگلیسی، ارقام، علائم نگارشی و نویسه‌های کنترلی

Reference: [Real python website](#)

تبدیل کاراکتر به عدد

تابع `ord()`

یک رشته شامل یک کاراکتر را دریافت کرده و مقدار عددی (صحیح) آن را برمی‌گرداند

```
In [1]: char_value = 'g'
        ascii_code = ord(char_value)
        print(ascii_code)
```

103

تبدیل عدد به کاراکتر

تابع `chr()`

یک عدد صحیح را دریافت کرده و رشته کاراکتری متناظر با آن را برمی‌گرداند

```
In [2]: ascii_code = 103
        char_value = chr(ascii_code)
        print(char_value)
```

g

تبدیل کل یک رشته به ASCII:

برای دریافت کدهای مقادیر تمام کاراکترهای یک رشته می‌توانید از

(list comprehension) استفاده کنید

```
In [5]: python_text = "Hello"
        ascii_list = [ord(char) for char in python_text]
        print(ascii_list)
```

[72, 101, 108, 108, 111]

Number guesser mini project

```
In [3]: import random

# random is a module that you can use all its function

# r = random.randrange(1,5) # this doesnt include stop, but use randint to include
# print(r)

top_of_range = input('Type a number: ')

# below condition will do this = '23' -> 23
if top_of_range.isdigit():
    top_of_range = int(top_of_range)
    if top_of_range <= 0:
        print('Please enter a number greater than 0')
        quit()
else:
    print('Please enter a number')
    quit()

random_number = random.randint(0, top_of_range)
guesses = 0
while True:
    guesses += 1
    user_guess = input('Guess a number: ')
    if user_guess.isdigit():
        user_guess = int(user_guess)
    else:
        print('Please enter a number')
        continue
    if user_guess == random_number:
        print('You got it!')
        break
    else:
        print('Sorry, you didn\'t get it.')
        if user_guess > random_number:
            print('Your guess is too high')
        else:
            print('Your guess is too low')

print(f'you got it in {guesses} guesses.')
```

Sorry, you didn't get it.
Your guess is too low
Sorry, you didn't get it.
Your guess is too high
Sorry, you didn't get it.
Your guess is too low
Sorry, you didn't get it.
Your guess is too low
Sorry, you didn't get it.
Your guess is too low
Sorry, you didn't get it.
Your guess is too low
You got it!
you got it in 7 guesses.